

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<tool_call> Reasoning process here :

Step 1 : Understand the total file size and normal download speed .

- Total file size = 200 GB
- Normal download speed = 2 GB / minute

Step 2 : Find out where the restart happens .

- 40 % of the way through the download $\hat{=}$ 40 % of 200 GB = $0.40 \times 200 = 80$ GB

Step 3 : Calculate time to download 80 GB at 2 GB / minute .

- Time = (80 GB) / (2 GB / minute) = 40 minutes

Step 4 : The restart happens and takes 20 minutes .

- Time for restart = 20 minutes

Step 5 : After restart , the download resumes from the beginning .

- So , the remaining 120 GB (200 - 80) must be downloaded from the start again at 2 GB / minute .
- Time to download remaining 120 GB = $120 / 2 = 60$ minutes

Step 6 : Add up all times .

- Time = 40 minutes (first part) + 20 minutes (restart) + 60 minutes (re-download)
- Total time = $40 + 20 + 60 = 120$ minutes

Therefore , the total time it takes Carla to download the file is 120 minutes .

< answer > 120 </answer >

